

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		$\frac{\text{speed of light in a vacuum}}{\text{speed of light in the material}}$ Do not accept $n = \frac{c}{v}$ unless letters are clearly defined. Accept answer based on ratio of sines if surrounding material stated as vacuum / $n = 1$ / accept air for vacuum	1			1		
	(b)	(i)	Substitution into: $n_1 \sin \theta_1 = n_2 \sin \theta_2$ i.e. $1.61 \sin 65 = n_2 \sin 80$ (1) $n_2 = 1.48$ (1)	1	1		2	2	
		(ii)	Distance travelled = $\frac{10}{\cos 80} = 58$ [mm] (1) $\frac{3 \times 10^8}{1.48 \text{ ecf}} = 2.03 \times 10^8$ [m s ⁻¹] (1) $\frac{58 \times 10^{-3}}{2.03 \times 10^8} = 2.86 \times 10^{-10}$ [s] (1)		3		3	3	
	(c)	(i)	$\sin C = \frac{1.31}{1.48 \text{ ecf}}$ (1) $C = 62.3^\circ$ (1) $< 80^\circ \therefore$ TIR (1) Accept $n_1 \sin \theta_1 = n_2 \sin \theta_2$ with 80° (1) Reports error message (1) Hence no refraction therefore TIR (1)	1 1	1		3	2	
		(ii)	Light reflected at Y with angle $i =$ angle r by eye		1		1		
			Question 5 total	4	6	0	10	7	0